

Six-leg calibration gate for the five trained families

a

	positives ≥ 3	held-out positive	explicit negatives	challenge rejected	family-resolved negatives	challenge status	verdict
hfam_70 · dPHASCL type 2	7 papers	1 (0.484)	2	10/10	sufficient	sufficient	PASS (all legs)
hfam_52 · dPHASCL type 1	~4 strong + 2 weak	1 (id 0.304)	2	10/10	sufficient	sufficient	PASS with caveat (2 weak positives)
hfam_55 · dPHASCL type 1	2 (<3)	1 (id 0.701)	2	10/10	sufficient	sufficient	NOT unlocked
hfam_8 · dPHAMCL	2 (<3)	1 (id 0.840)	2	10/10	sufficient	sufficient	NOT unlocked
hfam_4 · nPHAMCL	1 (<3)	leakage 0.901	1 (PcaD)	3 false positives	sufficient	not sufficient	NOT unlocked
37 reference-only families	no HMM built	no HMM built	no HMM built	no HMM built	no HMM built	no HMM built	NOT unlocked

b Hits on negative / confounder controls (lower = more specific)

Trained-family challenge (10 lipases/esterases + PcaD)

trained family HMMs



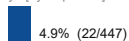
x1 non-hydrophobic confounder set

discovery layer HMM



Non-Cys reference superfamilies

Cys [hydrophobic]-C-Q motif



Positive control: the Cys [hydrophobic]-C-Q motif hits 271/276 = 98.2% of Cys references.

c Proactively flagged / isolated layers (never deleted)

x1 non-hydrophobic confounders flagged

kept as candidates, flagged not deleted

563

SignalP-predicted secretion conflicts demoted

0.18% of Cys; reversible, prediction-based

52

Pfam architecture conflicts held

hold_architecture_conflict

83

Ambiguous rows kept unassigned

not forced into any superfamily

616

Pool-external weak/mid not upgraded

score-only tier, excluded from all counts

4,926

with-lipase deferred layer preserved

known noise; deletion forbidden; outside all counts

1,206,655

Bar length is log10(count); the 1.2M deferred layer is preserved outside all counts.